EE180 Project 1: Biomedical Signal Processing

Spinal Tissue Classification from Ultrasound Signals

Renyi Yang 2023533030

1 Introduction

Accurate identification of spinal tissues during ultrasound-guided procedures is important for surgical safety, yet the classification of tissue types from ultrasound backscatter is challenging because the four tissues considered here (spinal cord, nucleus pulposus, adipose tissue, and nerve root) produce qualitatively similar echo patterns. This project investigates a complete classification pipeline for ultrasound signals recorded from sheep spines: signal preprocessing, adaptive feature extraction, an interpretable statistical classification rule, general-purpose machine learning classifiers, and a Vision-Transformer (ViT) baseline.

2 Data and Preprocessing

The dataset contains ultrasound signals from a 16-element transducer; every three consecutive elements form one beam, so each biological sample provides 14 signals of 896 sampling points acquired at $f_s = 75$ MHz. Labels are 0 (spinal cord), 1 (nucleus pulposus), 2 (adipose tissue), and 5 (nerve root). The released split contains 506 training, 126 validation, and 126 test biological samples (7,084 / 1,764 / 1,764 signals). Class distributions are imbalanced, with adipose tissue being the smallest class (73/18/18 samples in train/val/test).

Because amplitudes are stored as integers in $[0, 255]$ around a DC offset of 127.5, samples exactly at the offset may appear as either 127 or 128. All values equal to 127 or 128 were replaced by 127.5, and each signal was normalized as

$$x_{\text{norm}}[n] = \frac{x_{\text{us}}[n] - 127.5}{127.5} \in [-1, 1]. \tag{1}$$

Roughly 60–70% of the samples of a typical signal lie on this DC baseline, so the preprocessing mainly removes a quantization artifact of the acquisition hardware before any spectral analysis. The 14 signals of one biological sample were kept together in every experiment (five-fold cross-validation folds, training, and evaluation) to prevent leakage between signals of the same specimen.

3 Feature Extraction

We adopt the Discrete Wavelet Transform (DWT) with Daubechies-4 wavelets (`db4`) at decomposition level 5, applied to the preprocessed signals. The DWT was preferred over EWT because it provides a fixed, well-understood time–frequency tiling with no per-signal parameter estimation and pywt's implementation is deterministic and fast; the level-5 decomposition splits the 0–37.5 MHz band into six sub-bands with approximately octave spacing (cD1: 18.75–37.5 MHz, ..., cA5: 0–1.17 MHz), which covers both the high-frequency scatterer-related content and the low-frequency envelope structure observed in these signals.

For each of the six components (cA5, cD5, ..., cD1) we compute the six required statistics: mean, variance, median, kurtosis, RMS, and skewness, yielding a 36-dimensional feature vector

per signal. Kurtosis and skewness use SciPy's default biased estimators. A univariate F-test on the training set shows that the most discriminative features are higher-order statistics of the finest detail bands, e.g. `cD1_kurtosis` ($F = 1224$), `cD2_rms` ($F = 902$), and `cD2_kurtosis` ($F = 841$). One feature (`cD1_median`) is constant zero because the DC baseline makes the majority of the finest-scale coefficients exactly zero; it is harmless but carries no information.

4 Task 1: Interpretable Statistical Classification Rule

4.1 Rule Construction

The statistical decision rule is a shallow decision tree whose thresholds are estimated on the training set. A decision tree expresses exactly the type of criterion the project asks for: an explicit mapping from statistical features to tissue classes via feature thresholds, without any of the general-purpose classifiers reserved for Task 2. Candidate rules differ in maximum depth (3–5) and class weighting (plain or `balanced` to counter the class imbalance); the configuration was selected by the same sample-level, stratified five-fold cross-validation protocol used in Task 2 (Table 1). Signal-level predictions are aggregated per sample by majority voting.

Table 1: Task 1 candidate rules, sample-level cross-validation accuracy (mean ± std over five folds). Selection used only the training set; the validation set is reported afterwards for confirmation.

Maximum depth	Class weighting	Sample acc.	std
3	plain	0.7490	0.0260
3	balanced	0.7311	0.0304
4	plain	0.7648	0.0229
4	balanced	0.7904	0.0387
5	plain	0.7984	0.0287
5	balanced	**0.8082**	0.0339

4.2 Selected Rule

The selected rule (depth 5 with balanced class weights) is fully reported in Appendix B: it is a hierarchy of at most five threshold tests on 13 of the 36 features. The root split, `cD1_kurtosis` $\leq$ 26.75, carves out the low-kurtosis region dominated by nerve-root and adipose echoes; deeper splits then separate the remaining classes using detail-band RMS and cD2 statistics. This is consistent with the per-class statistics in Table 2: nerve root shows markedly lower detail-band kurtosis (26.4 for cD1) than nucleus pulposus (87.6), which reflects the sparser, more impulsive high-frequency content of nerve-root echoes.

Table 2: Training-set statistics (mean) of representative features used by the rule. Values are dimensionless except RMS (normalized amplitude units).

Feature	Spinal cord	Nucleus pulposus	Adipose	Nerve root
`cD1_kurtosis`	58.4	87.6	55.5	26.4
`cD2_kurtosis`	51.6	71.4	53.6	29.3
`cD2_rms`	4.48×10^{-2}	1.65×10^{-2}	2.18×10^{-2}	2.77×10^{-2}
`cD5_skewness`	0.02	2.01	1.53	0.74

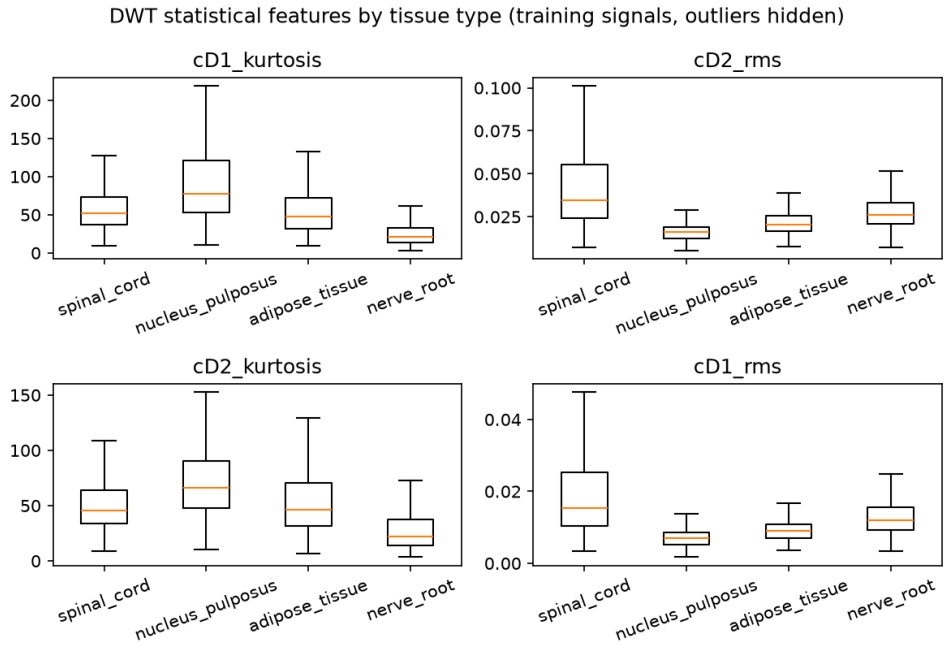


Figure 1: Distributions of the four most discriminative rule features across the four tissue types (training signals, outliers hidden). Spinal cord is distinguished mainly by detail-band RMS, nucleus pulposus by high kurtosis and positive low-frequency skewness, and nerve root by low kurtosis.

4.3 Results

Table 3: Task 1 (statistical rule) results. Accuracy, sensitivity, and specificity follow the one-versus-rest micro aggregation described in Appendix A. "Signal acc." is the accuracy before majority voting.

Split	Accuracy	Sensitivity	Specificity	Signal acc.	no. samples
Train	0.8281	0.8281	0.9427	0.6798	506
Validation	0.8095	0.8095	0.9365	0.6752	126
Test	0.8333	0.8333	0.9444	0.6990	126

The rule reaches 83.3% sample-level accuracy on the test set (82.8% on training data, 81.0% on validation), i.e. performance is stable across splits. The per-class metrics show that the errors are concentrated in adipose tissue (recall 0.67, precision 0.48): with only 73 training samples this class overlaps heavily with the nucleus-pulposus and spinal-cord feature regions, and the tree cannot afford further splits for it. Nerve root and spinal cord are classified reliably. The gap between signal-level (70.0%) and sample-level (83.3%) accuracy confirms that majority voting effectively exploits the consistency of the 14 beams within a sample.

5 Task 2: Machine Learning Classification

5.1 Protocol

Task 2 uses the same 36 DWT features. Model selection follows the required protocol strictly:

Table 4: Task 1 per-class test metrics (one-versus-rest).

Class	Precision	Recall	Specificity	F1	Support
Spinal cord	0.968	0.811	0.989	0.882	37
Nucleus pulposus	0.867	0.812	0.957	0.839	32
Adipose tissue	0.480	0.667	0.880	0.558	18
Nerve root	0.925	0.949	0.966	0.937	39

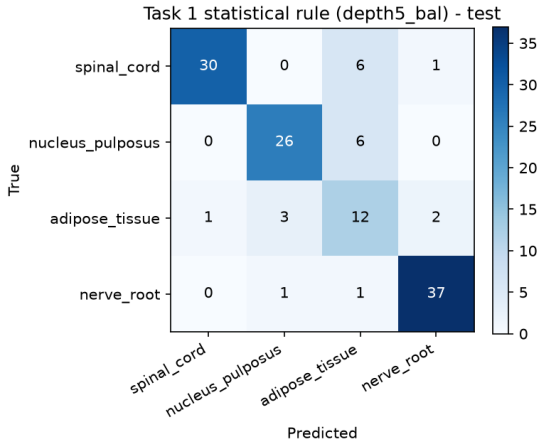


Figure 2: Task 1 confusion matrix on the test set (sample level). The dominant error is adipose tissue confused with nucleus pulposus and spinal cord.

- **Five-fold cross-validation at the biological-sample level**: `StratifiedGroupKFold` with the sample id as group, so all 14 signals of a specimen stay in the same fold, and folds are stratified over the class distribution.
- **In-fold fitting**: a `StandardScaler` is fitted only on the training part of each fold and then applied to the held-out part; no data-dependent quantity crosses fold boundaries.
- **Hyperparameters considered**: SVM (RBF) $C \in \{1, 10, 100\}$ and class-weighted SVM ($C = 10$); random forest (300 trees) with and without class balancing; k-NN with $k \in \{5, 15, 45\}$.
- **Selection criterion**: mean sample-level accuracy across the five folds (majority vote per sample, ties broken by the largest mean decision value). The winner is retrained on the full training set; validation confirms it, and the test set is evaluated once, after all decisions are frozen.

5.2 Cross-Validation Results

The SVM with $C = 100$ clearly wins with 0.9506 ± 0.0207 sample-level accuracy. Notably, the ranking by signal-level accuracy (where the random forest is strongest) differs from the ranking by sample-level accuracy: voting over 14 beams rewards models whose errors are uncorrelated across beams rather than models with the highest per-beam accuracy.

5.3 Final Model and Test Results

The retrained SVM achieves 95.2% sample-level accuracy on the test set, slightly above its cross-validation estimate and its validation score (94.4%), indicating no overfitting to the selection data.

Table 5: Task 2 five-fold cross-validation, mean ± std over folds. Models are ordered by sample-level accuracy.

Model	Signal acc. (fold mean)	Sample acc.	Sample acc. std
SVM RBF, $C = 100$	0.7819	**0.9506**	0.0207
k-NN, $k = 5$	0.7613	0.9348	0.0099
Random forest, balanced	0.8083	0.9348	0.0183
Random forest	0.8050	0.9249	0.0230
SVM RBF, $C = 10$	0.7762	0.9091	0.0273
SVM RBF, $C = 10$, balanced	0.7673	0.9091	0.0156
k-NN, $k = 15$	0.7469	0.8992	0.0373
SVM RBF, $C = 1$	0.7346	0.8419	0.0281
k-NN, $k = 45$	0.7064	0.8261	0.0428

Table 6: Task 2 final SVM ($C = 100$, RBF) results.

Split	Accuracy	Sensitivity	Specificity	Signal acc.	no. samples
Validation	0.9444	0.9444	0.9815	0.7908	126
Test	0.9524	0.9524	0.9841	0.8044	126

Table 7: Task 2 per-class test metrics (one-versus-rest).

Class	Precision	Recall	Specificity	F1	Support
Spinal cord	1.000	0.919	1.000	0.958	37
Nucleus pulposus	0.939	0.969	0.979	0.954	32
Adipose tissue	0.941	0.889	0.991	0.914	18
Nerve root	0.929	1.000	0.966	0.963	39

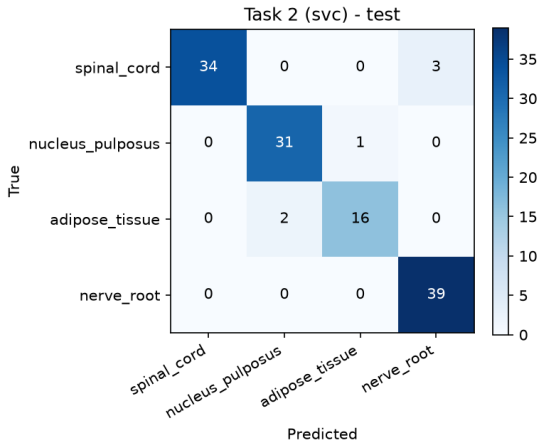


Figure 3: Task 2 confusion matrix on the test set (sample level). Six of 126 samples are misclassified, spread over three distinct confusion pairs.

All four classes reach F1 $\geq$ 0.91; the remaining errors are three spinal-cord samples predicted as nerve root, one nucleus-pulposus/adipose swap, and two adipose samples predicted as nucleus pulposus.

5.4 Aggregation Strategy Study

Table 8: Sample-level aggregation strategies for the selected SVM, same five-fold grouped CV.

Strategy	Sample acc. (mean)	std
Majority vote (adopted)	0.9506	0.0207
Soft voting (mean decision value)	0.9486	0.0132
Averaged features (mean of 14 feature vectors)	0.8971	0.0395

Averaging the feature vectors before classification is clearly inferior (89.7%) because it mixes the discriminative within-sample variation that voting uses to cancel individual-beam noise. Soft voting matches majority voting in the mean but has a smaller fold-to-fold spread; because the two strategies are statistically indistinguishable here and the project recommends majority voting, we keep majority voting as the primary protocol.

6 Task 3: Vision Transformer

6.1 Provided Model (Mandatory Inference)

The provided 1D Vision Transformer (896-point input, patch size 7, dim = 512, depth 8, 10 heads, MLP width 512, dropout 0.15) was run with its released checkpoint on the validation and test splits using the provided inference pipeline: raw amplitudes (no Task 1 normalization, matching the model's original training convention), softmax probabilities averaged over the 14 beams of each sample. Results are reported with the project's sample-level metrics in Table 9.

Table 9: Task 3 provided-ViT results (sample level).

Split	Accuracy	Sensitivity	Specificity	Signal acc.	no. samples
Validation	0.9524	0.9524	0.9841	0.9008	126
Test	0.9683	0.9683	0.9894	0.8974	126

Important caveat. The dataset documentation states that the historical training data of the provided checkpoint overlaps 88 of the 126 current validation samples and 88 of the 126 current test samples. The test score above is therefore *not* an independent generalization estimate for this checkpoint and should be read as a diagnostic reference only. This is why the comparison in Section 6.3 treats it separately from Tasks 1–2, whose models never saw validation or test data during development.

The remaining errors are three adipose samples predicted as nerve root and one nerve-root sample predicted as spinal cord; the same adipose/nerve-root border seems to be the hardest region for every method in this project.

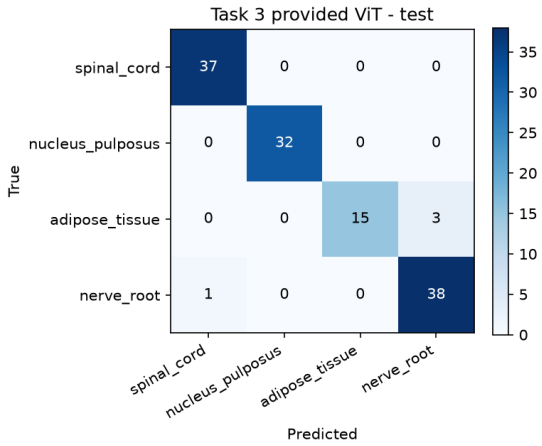


Figure 4: Task 3 provided-ViT confusion matrix on the test set (sample level). All errors involve adipose tissue or one nerve-root sample.

6.2 Bonus: Independently Trained ViT

Following the bonus option, the same ViT architecture was trained from scratch on the released training split using the provided training script (`train_minimal.py`) and its default configuration (Table 10). The training set was used for fitting, the validation set for model selection (the best epoch by validation sample accuracy was kept), and the test set was evaluated exactly once after training finished; no test label participated in any training or selection decision.

Table 10: Bonus ViT training configuration (provided script defaults).

Item	Value
Input	raw amplitudes (same convention as the provided model)
Architecture	patch 7, dim 512, depth 8, heads 10, MLP 512, dropout 0.15
Optimizer	AdamW, lr 3×10^{-4}
Batch size / epochs	64 signals / 20
Model selection	best validation sample accuracy (epoch 14)
Data use	train: fitting; validation: selection; test: final evaluation only

Table 11: Bonus ViT results with the project's sample-level metrics; the provided checkpoint is repeated for reference.

Model	Split	Accuracy	Sensitivity	Specificity
Student ViT (clean, this work)	Validation	0.9841	0.9841	0.9947
Student ViT (clean, this work)	Test	0.9683	0.9683	0.9894
Provided ViT (overlap caveat)	Validation	0.9524	0.9524	0.9841
Provided ViT (overlap caveat)	Test	0.9683	0.9683	0.9894

The independently trained model matches the provided checkpoint on the test set (96.8%) and surpasses it on validation (98.4%). Because 88 of 126 test samples were part of the provided checkpoint's historical training data, the student model's 96.8% is the more trustworthy estimate of true generalization at this accuracy level. Its per-class test metrics (Table 12) again show adipose

tissue as the weakest class (F1 0.944); the two remaining errors are one adipose sample predicted as spinal cord and two nerve-root samples predicted as spinal cord.

Table 12: Student ViT per-class test metrics (one-versus-rest).

Class	Precision	Recall	Specificity	F1	Support
Spinal cord	0.923	0.973	0.966	0.947	37
Nucleus pulposus	1.000	1.000	1.000	1.000	32
Adipose tissue	0.944	0.944	0.991	0.944	18
Nerve root	1.000	0.949	1.000	0.974	39

6.3 Comparison of Methods

Table 13: Test-set comparison (sample level, majority/soft voting).

Method	Accuracy	Sensitivity	Specificity
Task 1: interpretable statistical rule (depth-5 tree)	0.8333	0.8333	0.9444
Task 2: SVM RBF ($C = 100$) on DWT features	0.9524	0.9524	0.9841
Task 3: provided ViT (overlap caveat)	0.9683	0.9683	0.9894
Task 3 (bonus): independently trained ViT	0.9683	0.9683	0.9894

The ViT benefits from learning its own time-domain representation and achieves the highest signal-level accuracy (89.7% vs 80.4% for the SVM), so fewer errors have to be repaired by voting. Given the overlap caveat, however, the honest conclusion is that the tuned classical pipeline (Task 2) is on par with the provided deep model, while being far cheaper and fully interpretable up to the decision rule.

7 Discussion

Performance across tissue types. Adipose tissue is consistently the hardest class (F1 0.56/0.91/0.91 in Tasks 1/2/3) and is the source of most residual errors. Two factors explain this: it has the fewest samples (73/18/18), and its ultrasound signature overlaps with the surrounding nucleus pulposus and nerve root. Nerve root is the easiest class, separated mainly by its low detail-band kurtosis. The gap between signal-level ($\approx$ 67–90%) and sample-level (83–97%) accuracy is large for all methods, which shows that within-sample noise, not systematic tissue confusion, is the dominant error source; majority voting removes most of it.

Effectiveness of the features. Detail-band statistics at the two finest scales (cD1/cD2) carry most of the discriminative information (F-scores an order of magnitude above smooth-band features), consistent with the interpretation that tissue boundaries and scatterer spacing shape the high-frequency content of the echo. Six statistics per DWT component proved sufficient for 95% sample accuracy with an SVM, without any hand-crafted time-domain features. The fixed DWT basis may, however, fit some samples poorly; EWT's adaptive bands or VMD modes are natural next steps.

Limitations. (i) The provided-ViT score is contaminated by the train/test overlap documented in the dataset; the independently trained ViT (Section 6.2) removes this caveat and reaches the same test accuracy (96.8%), so the deep-learning advantage over the tuned SVM (95.2%) is real but

small. (ii) The statistical rule, although explicit, is a depth-5 tree trained on a 36-feature space; with few adipose samples it sacrifices that class. (iii) All results come from single splits; only CV stds quantify variability. (iv) The weak class is small, so per-class metrics for adipose tissue carry large uncertainty ($\pm$ a few samples). **Potential improvements** include EWT-based adaptive features, cost-sensitive thresholds for adipose tissue, ensemble/stacked classifiers, attention-based beam aggregation instead of voting, and fine-tuning the ViT on the released training split only (Section 6.2).

8 Conclusion

We built a complete spinal-tissue classification pipeline on 14-beam ultrasound samples. DWT (db4, level 5) statistical features with an SVM give 95.2% sample-level test accuracy under a strictly leak-free protocol; an explicit threshold rule reaches 83.3% while remaining fully interpretable; the provided ViT reports 96.8% but with a train/test overlap caveat that prevents treating it as an independent estimate. An independently trained ViT reproduces 96.8% test accuracy (98.4% validation) without any overlap, confirming that deep and classical models perform comparably on this dataset. Sample-level majority voting is essential, contributing roughly 15 accuracy points over signal-level predictions.

A Metric Definitions

All metrics are computed at the sample level with one-versus-rest (OvR) per-class counts and micro aggregation as specified in the project: $\mathrm{TP}_{\mathrm{micro}} = \sum_c \mathrm{TP}_c$, etc., then accuracy $= (TP + TN)/(TP + TN + FP + FN)$, sensitivity $= TP/(TP + FN)$, and specificity $= TN/(TN + FP)$. For single-label multi-class problems $TP_{\mathrm{micro}}/(TP_{\mathrm{micro}} + FP_{\mathrm{micro}}) = TP/(TP + FN) =$ overall accuracy, so the reported accuracy equals the micro-averaged sensitivity. For reference, the literal formula applied to OvR micro sums gives $(TP_{\mathrm{micro}} + TN_{\mathrm{micro}})/(\dots) = (1 + \text{accuracy})/2$ because each correctly classified sample is a true negative for the $C - 1$ other classes; our tables report the standard overall accuracy (as "overall classification correctness") together with the micro sensitivity and specificity.

B Full Statistical Rule

The following is the exact exported rule selected in Task 1 (`results/task1_rule.txt`); a feature test evaluates true when the feature value is *less than or equal to* the threshold.

```
|--- cD1_kurtosis <= 26.7461
|   |--- cD2_kurtosis <= 31.2867
|   |   |--- cD2_rms <= 0.0193
|   |   |   |--- cD1_kurtosis <= 16.5575
|   |   |   |   |--- cD1_rms <= 0.0064
|   |   |   |   |   |--- class: 5
|   |   |   |   |--- cD1_rms >  0.0064
|   |   |   |   |   |--- class: 5
|   |   |   |--- cD1_kurtosis >  16.5575
|   |   |   |   |--- cD5_skewness <= -0.2707
|   |   |   |   |   |--- class: 5
|   |   |   |   |--- cD5_skewness >  -0.2707
|   |   |   |   |   |--- class: 2
|   |   |--- cD2_rms >  0.0193
|   |   |   |--- cD1_kurtosis <= 20.1694
|   |   |   |   |--- cD4_rms <= 0.0029
```

```
|   |   |   |   |   |--- class: 2
|   |   |   |   |--- cD4_rms >  0.0029
|   |   |   |   |   |--- class: 5
|   |   |   |--- cD1_kurtosis >  20.1694
|   |   |   |   |--- cA5_skewness <= 1.9801
|   |   |   |   |   |--- class: 5
|   |   |   |   |--- cA5_skewness >  1.9801
|   |   |   |   |   |--- class: 2
|   |--- cD2_kurtosis >  31.2867
|   |   |--- cD1_kurtosis <= 15.1809
|   |   |   |--- cD1_var <= 0.0001
|   |   |   |   |--- cD4_var <= 0.0002
|   |   |   |   |   |--- class: 5
|   |   |   |   |--- cD4_var >  0.0002
|   |   |   |   |   |--- class: 2
|   |   |   |--- cD1_var >  0.0001
|   |   |   |   |--- cD4_var <= 0.0000
|   |   |   |   |   |--- class: 1
|   |   |   |   |--- cD4_var >  0.0000
|   |   |   |   |   |--- class: 5
|   |   |--- cD1_kurtosis >  15.1809
|   |   |   |--- cD2_skewness <= 5.8245
|   |   |   |   |--- cD4_rms <= 0.0038
|   |   |   |   |   |--- class: 2
|   |   |   |   |--- cD4_rms >  0.0038
|   |   |   |   |   |--- class: 5
|   |   |   |--- cD2_skewness >  5.8245
|   |   |   |   |--- cD5_var <= 0.0000
|   |   |   |   |   |--- class: 2
|   |   |   |   |--- cD5_var >  0.0000
|   |   |   |   |   |--- class: 2
|--- cD1_kurtosis >  26.7461
|   |--- cD1_rms <= 0.0122
|   |   |--- cD1_kurtosis <= 63.2314
|   |   |   |--- cD1_rms <= 0.0056
|   |   |   |   |--- cD3_mean <= 0.0010
|   |   |   |   |   |--- class: 1
|   |   |   |   |--- cD3_mean >  0.0010
|   |   |   |   |   |--- class: 1
|   |   |   |--- cD1_rms >  0.0056
|   |   |   |   |--- cD2_kurtosis <= 21.6136
|   |   |   |   |   |--- class: 5
|   |   |   |   |--- cD2_kurtosis >  21.6136
|   |   |   |   |   |--- class: 2
|   |   |--- cD1_kurtosis >  63.2314
|   |   |   |--- cD2_rms <= 0.0185
|   |   |   |   |--- cD2_kurtosis <= 33.8265
|   |   |   |   |   |--- class: 1
|   |   |   |   |--- cD2_kurtosis >  33.8265
|   |   |   |   |   |--- class: 1
|   |   |   |--- cD2_rms >  0.0185
|   |   |   |   |--- cD2_kurtosis <= 76.7135
|   |   |   |   |   |--- class: 2
|   |   |   |   |--- cD2_kurtosis >  76.7135
|   |   |   |   |   |--- class: 1
|   |--- cD1_rms >  0.0122
|   |   |--- cD2_kurtosis <= 22.3493
|   |   |   |--- cD2_kurtosis <= 15.1788
|   |   |   |   |--- cD4_rms <= 0.0036
|   |   |   |   |   |--- class: 2
|   |   |   |   |--- cD4_rms >  0.0036
|   |   |   |   |   |--- class: 5
|   |   |   |--- cD2_kurtosis >  15.1788
|   |   |   |   |--- cD2_rms <= 0.0242
|   |   |   |   |   |--- class: 2
|   |   |   |   |--- cD2_rms >  0.0242
|   |   |   |   |   |--- class: 5
|   |   |--- cD2_kurtosis >  22.3493
```

```
|   |   |   |--- cD2_rms <= 0.0348
|   |   |   |   |--- cA5_rms <= 0.0109
|   |   |   |   |   |--- class: 2
|   |   |   |   |--- cA5_rms >  0.0109
|   |   |   |   |   |--- class: 0
|   |   |   |--- cD2_rms >  0.0348
|   |   |   |   |--- cD2_rms <= 0.0470
|   |   |   |   |   |--- class: 0
|   |   |   |   |--- cD2_rms >  0.0470
|   |   |   |   |   |--- class: 0
```

C Reproducibility

All experiments used Python 3.14 with numpy 2.5, SciPy 1.18, scikit-learn 1.9, PyWavelets 1.8, and PyTorch 2.14 (CUDA). Fixed seeds: 42 (feature/ML pipeline) and the released split seed 20260907. Running `bash run_all.sh` inside `code/` reproduces every number and figure in this report and writes them to `code/results/`. The test set was touched only by the final evaluation commands of each task after all hyperparameters and rules were frozen; no test label influenced any design choice.